

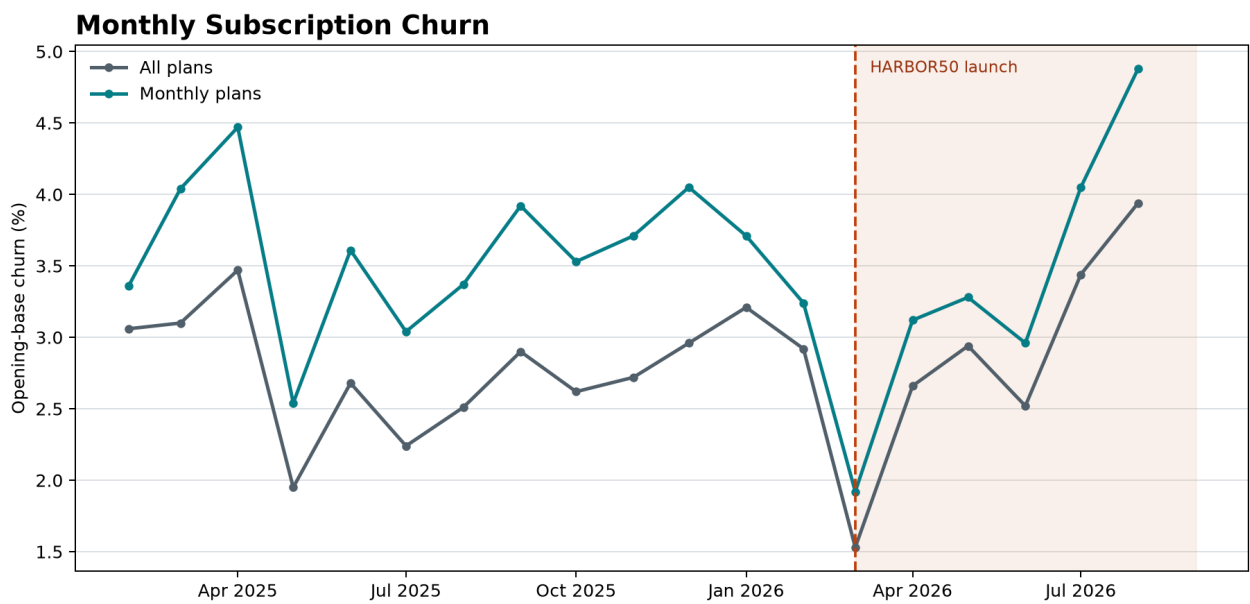
Recommendation

To: Dana Whitfield, VP Operations
Subject: August churn increase and HARBOR50

Churn Clarification

The available data supports a real increase in subscription churn from June to August, but the reported 2.7% and 5% are not directly comparable - it appears the board deck included all subscription plans while Finance is only including monthly plans; we should confirm this with each owner. My recommendation is to define monthly churn as the percentage of subscriptions active at the start of the month that cancel during the month. This definition excludes new in-month subscriptions and while it does increase from June to August, 2.5% to 3.9%, this is less alarming than the initial numbers from Finance suggest.

A significant contributor in the recent rise in churn is due to the expiration of the HARBOR50 discount - customers that took advantage of the promotion in March began paying full price in June, which is precisely when churn began to rise.



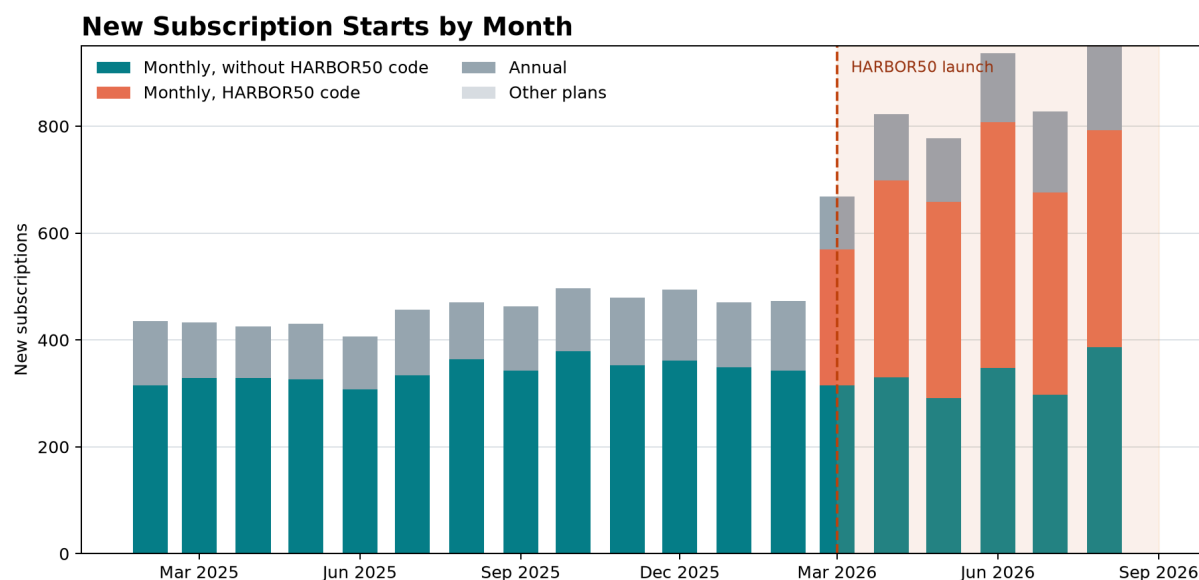
Cancellations among subscriptions active before the first of each month; September excluded (partial data).

HARBOR50 Promotion Performance

March-May cohorts retained 89.9% at day 90 in 2025 and 91.9% in 2026. However, HARBOR50 subscribers show a clear rise in churn around promotion expiry: the March cohort went from

2.1% in the 30 days before expiry to 13.3% in the next 30 days. April went from 2.6% to 12.6%, and the following 30-day window remains elevated: 20.8% and 17.8%, respectively. This is a signal worth investigating. Paid-social monthly subscribers without the code show a similar increase, though have a smaller sample size; from days 90-120 churn rose from 3.6% to 11.1% for March subscriptions and 8.6% to 12.5% for April. The rise may reflect a broader paid-social or month-four retention issue.

I recommend tracking 120- and 180-day retention and acquisition costs before making a decision to cancel the program. If feasible, use a randomized holdout or staged offer test, with a pre-agreed retention and contribution-margin threshold. Even if these customers do indeed cancel more frequently, the additional revenue for the discount period might outweigh the acquisition cost.



Questions for Dana

1. Can we confirm Marketing and Finance are computing churn in the same way I identified?
2. Can we start collecting information regarding why customers are cancelling subscriptions going forward? It seems that the promotional window expiring might be a large driver of churn, but the spike also occurred when inflation rose and forecasts for future interest rates increased. It is possible that customers left because our advisors didn't address these concerns in their interactions with customers.
3. 72 subscriptions have 'status = canceled' but no 'cancel_date'; can the missing event dates be recovered?
 - a. Does a cancellation date represent the request date, service end date, or billing stop date? How should same-day start/cancel events be treated?

4. Should COMP100 subscriptions be included in the company's paid churn KPI, and can dates be recovered for complimentary subscriptions with missing cancellation dates?
5. What are the expected post-discount price and renewal behavior for HARBOR50, and what retention or contribution-margin threshold would justify stopping it?
6. Can the next promotion wave include a randomized holdout or phased rollout?
7. Do we have an internal method for attributing channel spending to individual customers? This would assist in creating a method for measuring customer profitability. If not, we could estimate the average cost using total channel spend over new customers.

Appendix

Customer Satisfaction

Customer satisfaction remains consistent this year. Those responding 'yes' to our satisfaction survey make up 80.4, 81.8, and 81.2% of valid responses for Q1-Q3 of 2026, respectively.

Customer Growth

New subscriptions increased materially versus the same calendar month in 2025 and are largely driven by the HARBOR50 promotion.

Month	2025 monthly starts	2026 monthly starts	Change vs. 2025	HARBOR50-coded starts
March	329	569	+240	254
April	329	698	+369	368
May	326	658	+332	367
June	308	807	+499	460
July	334	676	+342	379
August	364	792	+428	406

These are observed year-over-year differences, not a causal estimate of added subscriptions. Meta spend rose sharply at the same time, and this dataset has neither a holdout nor campaign-level attribution; some code users may have subscribed without the discount.

Complimentary Subscriptions and Segmentation Churn

COMP100 promotion does not explain the June-to-August churn rise.

Among monthly subscriptions excluding COMP100, churn rose across every age band and region between June and August. The largest age-band rate increase was for ages 45–54 (2.53% to

5.14%, +2.61 percentage points); the largest regional increase was in the Southwest (2.18% to 5.42%, +3.23 points). The increase is broad, not isolated to one segment. Counts and opening denominators are in 'outputs/churn_by_age_region.csv'; these are unadjusted descriptive cuts, not evidence that age or region caused the change.

Revenue Estimation

For the HARBOR50 promotion, the maximum three-invoice concession is \$88.50 per subscriber before accounting for early cancellations. A SQL model estimates the next 12 months of full-price billings at \$549.53, \$465.02, and \$363.14 per acquired subscriber. Those post-promotion gross billings are about 6.4x, 5.5x, and 4.5x the expected concession, respectively.

Assumed monthly churn	Next 12 months at full price	Expected discount concession	Promoted 15-month gross billings	Full-price 15-month comparator
3%	\$549.53	\$85.87	\$635.40	\$721.27
5%	\$465.02	\$84.15	\$549.17	\$633.32
8%	\$363.14	\$81.61	\$444.75	\$526.36

These are scenario estimates, not observed revenue or profit. The model assumes monthly invoices at signup and each monthly anniversary, a constant churn hazard across the 15-month projection, three discounted invoices for subscribers still active, then full-price invoices. It excludes refunds, taxes, payment failures, variable service/support costs, and customer-level campaign attribution.

Definitions and Assumptions

- Opening base: subscriptions with 'start_date' strictly before the first of the month and no cancellation date on or before that date. A start or cancellation dated on the first is excluded from the opening snapshot.
- Opening-cohort churn: valid cancellations strictly after the first and before the next month's first, among opening subscriptions, divided by opening subscriptions. Cancellation dates before the start date are excluded as invalid. Because dates have no times, same-day ordering cannot be recovered; this convention avoids counting events on the snapshot boundary.
- Tenure retention: a subscription is retained at a checkpoint if its cancellation date is null or later than the checkpoint date. Only subscriptions whose checkpoint falls on or before September 15 are included.
- Cohort date: subscription 'start_date', not customer 'signup_date'.
- Promotion comparison: monthly subscriptions starting on or after March 1. 'HARBOR50' is confirmed via 'discount_code'; post-launch paid-social subscriptions without a code are labeled

likely campaign exposure, not confirmed offer users or non-users. Channel aliases are normalized for grouping. Results are observational, not causal.

- Complimentary subscriptions: 'COMP100' rows are kept separate from paid comparison groups. Churn sensitivity is reported both including and excluding them; HARBOR50 discounted subscribers remain in the paid population.

Data Quality Notes

- The files contain 11,461 customers and 11,681 subscriptions, with no duplicate customer or subscription IDs and no unmatched subscriptions in the customer join.
- 25 subscriptions have cancellation dates before their start dates; these records are excluded from cancellation-event and tenure-retention calculations.
- There are 120 COMP100 monthly subscriptions; 83 have 'status = canceled', and 38 have no cancellation date.
- 72 subscriptions are marked 'canceled' but have no cancellation date; none of the 'active' subscriptions have a cancellation date. 'cancel_date' for events and does not treat current status as a historical event date.
- 192 customers have a missing acquisition channel. They remain in overall metrics and are grouped as 'Unknown' in the channel analysis.
- 220 customers have multiple subscription rows. Subscription-level churn intentionally counts each subscription separately.
- Marketing spend is aggregated by channel and month; it cannot be joined to individual customers for causal acquisition or return-on-ad-spend estimates.

Reproducibility

From this folder, install the dependencies with 'py -m pip install -r requirements.txt', then run 'py create_charts.py'. This runs analysis.sql through 'run_analysis.py', regenerates the CSV outputs, and creates the three PNGs under 'outputs/charts/'. The script reads the raw survey CSV directly and saves its quarterly summary under 'outputs/quarterly_survey_recommendation.csv'.

Use of AI

I began looking into the data files with a Jupyter notebook and a python EDA script before prompting VS Code's AI agent to assist with planning my analysis. The AI helped set up DuckDB so I could interact with the tables easily, and structured the analysis and identified metric and data-quality edge cases. I executed the SQL locally and checked the reported row counts, June/August numerators and denominators, and cohort rates against the supplied CSVs.

Some corrections and clarifications I prompted:

- Focus the analysis on comparing those with the HARBOR50 discount vs. other subscribers
- Adjust the date filtering to not include the 1st of each month in the churn and retention calculations
- Check retention post-90 days to measure the impact
- Estimate revenue impact of promotion
- Inclusion of charts for key metrics
- Widen impact window from just March to March-May to increase volume of comparison group